# GUTAR GU — ZERO-TRACE EPHEMERAL CHAT

Complete Technical Implementation & AWS Cloud Deployment Plan (Free Tier + Demo Domain)

**Core Architecture Promise:** Gutar GU provides private, ephemeral 1-on-1 real-time chat with **client-side End-to-End Encryption (ECDH P-256 + AES-GCM-256)**, timed auto-destruct, and immediate zero-trace wiping across both devices upon user exit. Messages and keys exist solely in memory and never touch disks or databases.

**Deployment Objective:** Deploy on **AWS 100% Free Tier (or $100 Credits)** utilizing a free dynamic demo domain (**nip.io**) that automatically secures free **Let's Encrypt TLS/SSL certificates** without requiring purchase of a custom domain.

## 1. Core Feature Specification & Lifecycle

| Feature | Behavior & Privacy Guarantee | Mechanism |
|---|---|---|
| **E2EE (End-to-End Encryption)** | Zero plaintext on server. Server only relays opaque cipher payload. | `Web Crypto API (ECDH P-256 + AES-GCM-256)` |
| **Both-Ends Session Wipe** | When either participant disconnects or leaves, all DOM nodes shred and server memory purges instantly. | `WebSocket leave broadcast + CSS shredOut animation + room.clear()` |
| **Burn After Reading** | Vanishes immediately once partner renders and acknowledges read receipt. | `message_seen event + auto local destruction timer` |
| **Configurable Auto-Destruct** | Messages automatically self-destruct after 10s, 30s, 1m, or 5m. | `setTimeout + purge_message WS event` |
| **Ephemeral Media Sharing** | Peer-to-peer image transfer (<5MB) with tap-to-reveal view-once option. Never saved to disk. | `Base64 WebSocket frame relay + CSS blur filter` |

## 2. AWS Cloud Infrastructure Architecture

| Component | Free Tier / Credits Configuration | Zero-Trace Security Role |
|---|---|---|
| **Compute (EC2)** | t2.micro or t3.micro (Ubuntu 24.04 LTS) 750 hrs/month Free Tier | Runs FastAPI application entirely in RAM. Linux swap disabled (swapoff -a). |
| **Static Public IP** | AWS Elastic IP (100% Free while attached to EC2) | Stable endpoint for routing without leaking internal network identifiers. |
| **Demo Domain (No Cost)** | Wildcard DNS: <Elastic-IP>.nip.io | Maps directly to server IP with zero registration or payment required. |
| **TLS / SSL Reverse Proxy** | Caddy Server (Dockerized, Ports 80 & 443) | Auto-fetches Let's Encrypt certificates. Terminates HTTPS/WSS natively. |

# 3. Step-by-Step Deployment Instructions (From Windows)

Follow these 5 simple steps directly from your Windows web browser. No complex tools required.

### Step 1: Launch Free Tier EC2 Server on AWS

1. Log into your AWS Console and search for **EC2** → click **Launch Instance**.
2. **Name:** gutar-gu-server.
3. **OS:** Select **Ubuntu** (Ubuntu 24.04 or 22.04 LTS — marked *Free tier eligible*).
4. **Instance Type:** Select t2.micro (or t3.micro).
5. **Firewall / Security Group:** Check the boxes for:
   - ■■ **Allow SSH traffic** (Port 22)
   - ■■ **Allow HTTPS traffic from the internet** (Port 443)
   - ■■ **Allow HTTP traffic from the internet** (Port 80)
6. Click **Launch Instance**.

### Step 2: Allocate & Attach a Free Elastic IP

1. In the EC2 left sidebar under *Network & Security*, click **Elastic IPs**.
2. Click **Allocate Elastic IP address** → click **Allocate**.
3. Select the allocated IP, click **Actions** → **Associate Elastic IP address**.
4. Choose your gutar-gu-server instance and click **Associate**.
5. **Copy your Public IPv4** (e.g. 3.85.120.45).

### Step 3: Open Browser Terminal (EC2 Instance Connect)

1. In the EC2 Console, select your instance in the list.
2. Click the **Connect** button in the top menu bar.
3. Select **EC2 Instance Connect** and click **Connect**.
4. A terminal window immediately opens directly inside your web browser!

### Step 4: Clone Repository & Run Automated Setup

Inside your browser terminal, paste and run these commands:

```
sudo apt update && sudo apt install -y git
git clone https://github.com/shreyashhiray16/gutar-gu.git
cd gutar-gu/Gutar\ GU/chat-app
chmod +x setup_lightsail_ec2.sh
./setup_lightsail_ec2.sh
```

*The script will automatically detect your Elastic IP, configure the free <IP>.nip.io demo domain, install Docker, disable Linux swap memory, and bring up the Caddy reverse proxy with free Let's Encrypt SSL.*

### Step 5: Verify Live Deployment

Open your web browser and navigate to:
`https://<YOUR-ELASTIC-IP>.nip.io`
Confirm the padlock icon is active, test message exchange between two devices, and verify zero-trace session wiping on exit.

## 4. Zero-Trace Privacy & Security Hardening

• **No IP or Query Logging:** Access logging is disabled across Uvicorn and Caddy reverse proxy to prevent client IP retention.

• **In-Memory Execution Only:** No database engine (PostgreSQL/MySQL/MongoDB) is connected. Room objects live strictly in Python RAM.

• **Swap Protection:** Linux swap is completely disabled (swapoff -a) to ensure ephemeral data never flushes to server SSD.

• **Strict Content Security Policy (CSP):** Header restricts script execution and denies iframes (X-Frame-Options: DENY).

• **Client-Side Crypto Destruction:** Non-exportable CryptoKey references are destroyed instantly upon leave event.